

FP1 Chapter 6 Proof By Induction

Lecture 6 Proof By Induction

(Induction, Summation, Inequalities, Divisibility, nth Derivative)

Mathematical Induction

- The method of mathematical induction is useful in establishing the validity of general results or theorems which admit of successive cases corresponding to the numbers 1, 2, 3, ... n .
- The method is only applied to prove a stated result or theorem or one which can be conjectured from special cases.

The Method of Mathematical Induction

- We take n and k to denote positive integers and $n = 1$ to correspond to the simplest case. The procedure is as follows:
 - i. Verify that the proposed result or theorem in the simplest case, i.e. for $n = 1$.
 - ii. Assume that the result is true for $n = k$ and *prove* that it is true for $n = k + 1$.
 - iii. Combining (i) and (ii) we argue as follows:

From (i) the result is true for $n = 1$; hence follow (ii) it is true for $n = 2$. Again since the result is true for $n = 2$, it follows from (ii) that it is true for $n = 3$, and so on. Since the reasoning holds for any value of k of n it is quite general so that we may say that result is true for all finite integral values of n .

- Sometimes the statement is given, but we can conjecture / guess a proposition by verifying a few simple cases then proving it using mathematical induction. For example,

$$\frac{d}{dx}(xe^x) = xe^x + 1 \cdot e^x = (x + 1)e^x$$

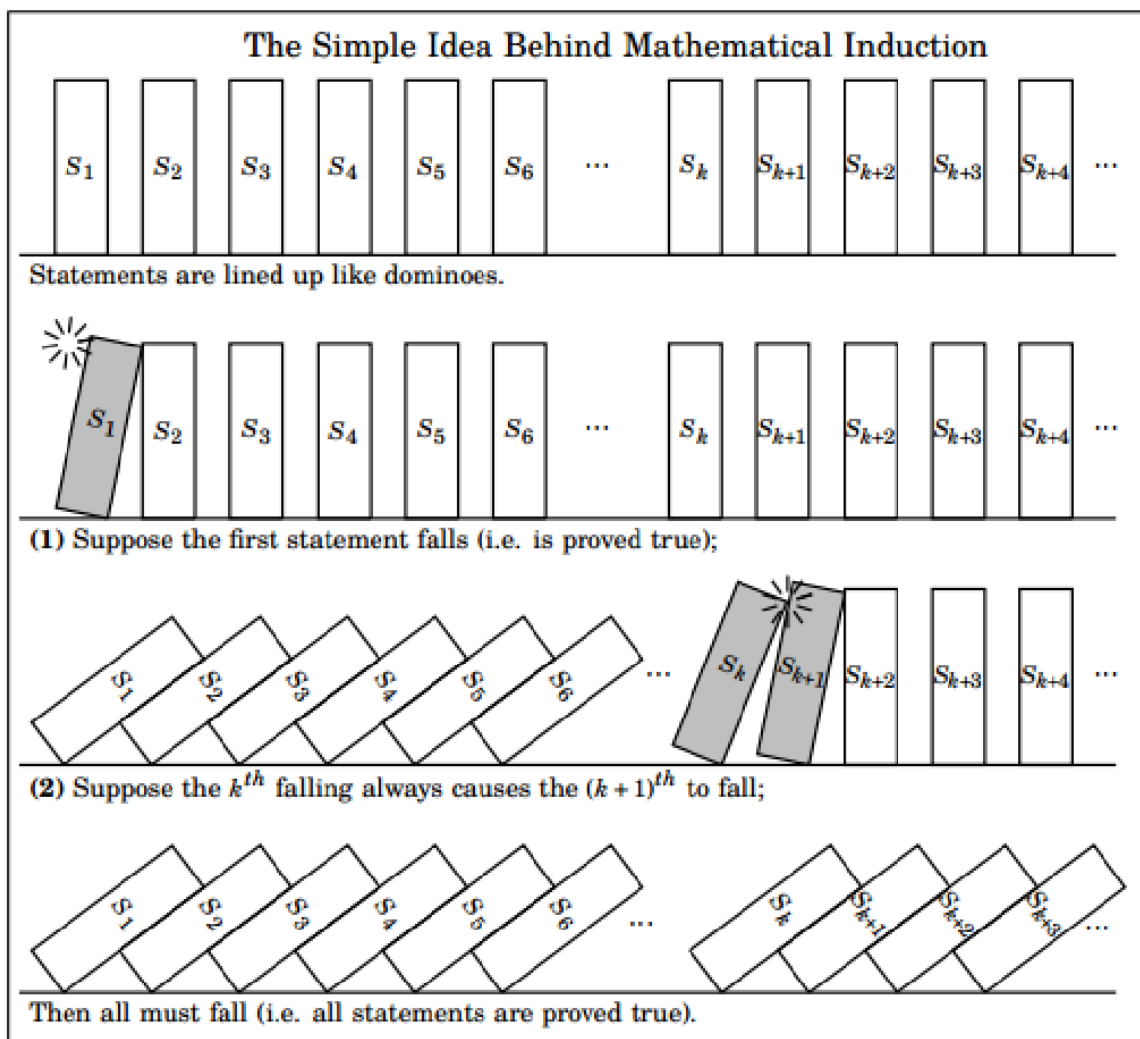
$$\frac{d^2}{dx^2}(xe^x) = \frac{d}{dx}[(x + 1)e^x] = (x + 1)e^x + 1 \cdot e^x = (x + 2)e^x$$

$$\frac{d^3}{dx^3}(xe^x) = \frac{d}{dx}[(x + 2)e^x] = (x + 2)e^x + 1 \cdot e^x = (x + 3)e^x$$

In this case, we can make a conjecture that

$$\frac{d^n}{dx^n}(xe^x) = (x+n)e^x.$$

- Notice that this method can only work for positive integers.
- The induction method can be illustrated as below.



Applications of the Method:

(1) Summation of Series

To prove that it is true for $n = k + 1$, we need to use the result

$$\sum_{r=1}^{k+1} u_r = \sum_{r=1}^k u_r + u_{k+1}.$$



- **Example 1:**

Prove by induction that

$$\sum_{r=1}^n (r^2 + 1)r! = n[(n + 1)!].$$

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- **Example 2:**

Prove by induction that

$$\sum_{n=1}^N n^3 = \frac{1}{4}N^2(N+1)^2.$$

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- **Example 3 (PYQ M/J 2017/13):**

Prove, by mathematical induction, that $\sum_{r=1}^n r \ln \left(\frac{r+1}{r} \right) = \ln \left(\frac{(n+1)^n}{n!} \right)$ for all positive integers n . [6]

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- **Example 4 (PYQ O/N 2012/13):**

- 3 Let $S_N = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{N}{(N+1)!}$. Prove by mathematical induction that, for all positive integers N ,

$$S_N = 1 - \frac{1}{(N+1)!}. \quad [5]$$

(2) Theorems on Divisibility

- Consider $f(n+1) - kf(n)$, where k is any integer that is not a multiple of the number you want to prove it is divisible by.
- For example, if we want to prove that $f(n)$ is divisible by 11, we can try $f(n+1) + f(n)$ or $f(n+1) - 3f(n)$, but trying $f(n+1) - 11f(n)$ would not help.
- **Example 5 (PYQ M/J 2006):**

Prove by induction, or otherwise, that

$$23^{2n} + 31^{2n} + 46$$

is divisible by 48, for all integers $n \geq 0$.

[6]

- **Example 6:**

Prove by induction that $3^{4n-2} + 17^n + 22$ is divisible by 16 for every positive integer n .

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- **Example 7:**

Prove by induction that $10^{3n} + 38^n + 35$ is divisible by 37 for every non-negative integer n .

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- **Example 8:**

Given that n is a positive integer, prove by induction that $5^{2n+2} - 24n - 25$ is a multiple of 576.

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- **Example 9:**

Prove by induction that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for all positive integers n .

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(3) Nth Derivative of a Function**- Example 10 (PYQ O/N 2015/11/12/13):**

Given that a is a constant, prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n}{dx^n}(xe^{ax}) = na^{n-1}e^{ax} + a^nxe^{ax}. \quad [6]$$

- **Example 11:**

If $y = e^{-x} \sin(x\sqrt{3})$, prove by induction that

$$\frac{d^n y}{dx^n} = (-2)^n e^{-x} \sin\left(x\sqrt{3} - \frac{1}{3}n\pi\right).$$

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- **Example 12 (PYQ O/N 2013/11/12):**

It is given that $y = (1+x)^2 \ln(1+x)$. Find $\frac{d^3y}{dx^3}$. [3]

Prove by mathematical induction that, for every integer $n \geq 3$,

$$\frac{d^n y}{dx^n} = (-1)^{n-1} \frac{2(n-3)!}{(1+x)^{n-2}}. \quad [5]$$

(4) Evaluation of Terms Defined by a Recurrence Relation

- A sequence can be defined by its n th term or by a recurrence relation.
- A recurrence relation is of the form

$$u_1 = \text{a given integer, 1 for example}$$

$$u_{n+1} = f(u_n)$$

- **Example 13:**

A sequence of positive numbers $u_1, u_2, u_3, \dots$ is defined by $u_1 = 1$ and $u_{n+1} = 3u_n + 2$ for $n \geq 1$.

Prove by induction that $u_n = 2(3^{n-1}) - 1$.

- **Example 14:**

A sequence of numbers $u_1, u_2, u_3, \dots$ is such that $u_1 = u_2 = 1$ and $u_{n+2} = u_{n+1} + u_n$ for $n \geq 1$. Prove that for all $n \geq 1$,

$$u_n = \frac{(1 + \sqrt{5})^n - (1 - \sqrt{5})^n}{2^n \sqrt{5}}.$$

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- **Example 15 (PYQ M/J 2012/13):**

For the sequence $u_1, u_2, u_3, \dots$, it is given that $u_1 = 1$ and $u_{r+1} = \frac{3u_r - 2}{4}$ for all r . Prove by mathematical induction that $u_n = 4\left(\frac{3}{4}\right)^n - 2$, for all positive integers n . [5]

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(5) Inequalities

Note that to prove that $a > b$, we might need to consider $a - b$ and prove that $a - b$ is positive.

- Example 16:

The sequence of real numbers $u_1, u_2, u_3, \dots$ is such that $u_1 = 1$ and $u_{n+1} = \frac{5u_n + 4}{u_n + 2}$ for all $n \geq 1$. Prove by induction that $u_n < 4$ for all $n \geq 1$.

- **Example 17:**

The sequence of positive numbers $u_1, u_2, u_3, \dots$ is defined by the relation

$$u_{n+1} = \frac{u_n^2 + 3u_n}{u_n + 1}$$

where $u_1 = 1$. Prove by induction that $u_n < 2n$ for all positive integers n .

- **Example 18:**

The sequence of numbers $u_1, u_2, u_3, \dots$ is defined by $u_{n+1} = 1 + \sqrt{u_n + 5}$, where $u_1 = 1$, prove by induction that $3 < u_n < 4$, for all $n \geq 2$.

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- **Example 19:**

Prove that, if the statement $n! > 2n + 2^n$ is true for $n = k$, where k is a positive integer, then it is true for $n = k + 1$.

Hence find the set of values of n for which the statement is true.

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- **Example 20 (PYQ M/J 2012/11/12):**

Prove, by mathematical induction, that, for integers $n \geq 2$,

$$4^n > 2^n + 3^n.$$

[5]

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- **Example 21 (PYQ M/J 2007):**

The sequence $x_1, x_2, x_3, \dots$ is such that $x_1 = 1$ and

$$x_{n+1} = \frac{1 + 4x_n}{5 + 2x_n}.$$

Prove by induction that $x_n > \frac{1}{2}$ for all $n \geq 1$. [5]

Prove also that $x_n > x_{n+1}$ for all $n \geq 1$. [3]

- **Example 22 (PYQ O/N 2004):**

The sequence of real numbers $a_1, a_2, a_3, \dots$ is such that $a_1 = 1$ and

$$a_{n+1} = \left(a_n + \frac{1}{a_n}\right)^\lambda,$$

where λ is a constant greater than 1. Prove by mathematical induction that, for $n \geq 2$,

$$a_n \geq 2^{g(n)},$$

where $g(n) = \lambda^{n-1}$.

[6]

Prove also that, for $n \geq 2$, $\frac{a_{n+1}}{a_n} > 2^{(\lambda-1)g(n)}$.

[3]

(5) Matrices**- Example 23 (PYQ M/J 2011/13):**

Let $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$. Prove by mathematical induction that, for every positive integer n ,

$$\mathbf{A}^n = \begin{pmatrix} 2^n & 3(2^n - 1) \\ 0 & 1 \end{pmatrix}. \quad [5]$$

- **Example 24:**

Prove by induction that, for all positive integers n ,

$$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{r}x^r + \cdots + x^n.$$

The result

$$\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r},$$

if used in your proof, must be justified.

Deduce that

$$1 - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^r \binom{n}{r} + \cdots + (-1)^n = 0.$$

- **Example 25 (PYQ O/N 2016/11/12/13):**

Using factorials, show that $\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$. [2]

Hence prove by mathematical induction that

$$(a+x)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}x + \dots + \binom{n}{r}a^{n-r}x^r + \dots + \binom{n}{n}x^n$$

for every positive integer n . [4]

- (6) Write down a conjecture based on a limited trial and followed by inductive proof

- **Example 26:**

Given that $y = xe^x$, write down expressions for $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ in terms of x .

Write down a conjecture for the corresponding expression for $\frac{d^ny}{dx^n}$.

Use induction to establish an expression for $\frac{d^ny}{dx^n}$.

- **Example 27 (PYQ 2003 O/N):**

Given that $y = x \sin x$, find $\frac{d^2y}{dx^2}$ and $\frac{d^4y}{dx^4}$, simplifying your results as far as possible, and show that

$$\frac{d^6y}{dx^6} = -x \sin x + 6 \cos x. \quad [3]$$

Use induction to establish an expression for $\frac{d^{2n}y}{dx^{2n}}$, where n is a positive integer. [5]

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- **Example 28 (PYQ 2014 O/N/11/12/13):**

It is given that $u_r = r \times r!$ for $r = 1, 2, 3, \dots$. Let $S_n = u_1 + u_2 + u_3 + \dots + u_n$. Write down the values of

$$2! - S_1, \quad 3! - S_2, \quad 4! - S_3, \quad 5! - S_4. \quad [2]$$

Conjecture a formula for S_n . [1]

Prove, by mathematical induction, a formula for S_n , for all positive integers n . [4]

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- **Example 29 (PYQ 2004 M/J):**

It is given that

$$\frac{d^n}{dx^n} \left(\frac{\ln x}{x} \right) = \frac{a_n \ln x + b_n}{x^{n+1}},$$

where a_n and b_n depend only on n .

(i) Find a_1 , a_2 and a_3 . [3]

(ii) Use mathematical induction to establish a formula for a_n . [5]